1. E(X_1) = 2, E(X_2) = 5, Var(X_1) = 1, Var(X_2) = 9, Corr(X_1, X_2)=0.5. What's the covariance between X_1 and X_2? What's the value of E($X_1 \cdot X_2$)? What's the variance of $3X_1 + X_2$? What's the covariance between $3X_1 + X_2$ and X_1?

2. Consider a consumer who can spend her money either on 4 consumptions C_0, $C_{1,1}$, $C_{1,2}$, $C_{1,3}$ or throw the money away. The prices of the 4 goods are P_0, $P_{1,1}$, $P_{1,2}$, $P_{1,3}$, respectively. She can't spend more money than her total wealth Y.

Her utility function is:

$U(C_0, C_{1,1}, C_{1,2}, C_{1,3}) = u(C_0) + \pi_{1,1} \cdot u(C_{1,1}) + \pi_{1,2} \cdot u(C_{1,2}) + \pi_{1,3} \cdot u(C_{1,3})$

where $\pi_{1,1}+\pi_{1,2}+\pi_{1,3}=1$ (this constraint is not important, you can

Q1: Write out her budget constraint.

Q2: Use the first order condition to determine how the relative prices $P_{1,1}/P_0$, $P_{1,2}/P_{1,1}$, $P_{1,3}/P_{1,1}$ are related to the consumptions at the optimal level? What's the economic interpretation of the relationships?

Q3: Based on the same set of first order conditions as of Q2, how the relatives $P_{1,1}/P_0$ and $P_{1,2}/P_{1,1}$, $P_{1,3}/P_{1,1}$ are related to the parameters $\pi_{1,1}, \pi_{1,2}, \pi_{1,3}$?